

Load the Dataset

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("students(2).csv")

print(df.head())
```

	Student_ID	Study_Hours	Final_Marks
0	1	1	35
1	2	2	40
2	3	3	45
3	4	4	50
4	5	5	55

Display Dataset Information

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 3 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Student_ID      20 non-null    int64
1   Study_Hours     20 non-null    int64
2   Final_Marks     20 non-null    int64
dtypes: int64(3)
memory usage: 612.0 bytes
None
```

Display Statistical Summary

```
print(df.describe())
```

	Student_ID	Study_Hours	Final_Marks
count	20.00000	20.000000	20.000000
mean	10.50000	8.050000	77.750000
std	5.91608	4.058454	44.143158
min	1.00000	1.000000	5.000000
25%	5.75000	5.000000	48.750000
50%	10.50000	8.000000	72.500000
75%	15.25000	11.250000	96.250000
max	20.00000	15.000000	170.000000

Compare the Two Variables

- Study_Hours
- Final_Marks

```
print(df.columns)
print(df.head())
```

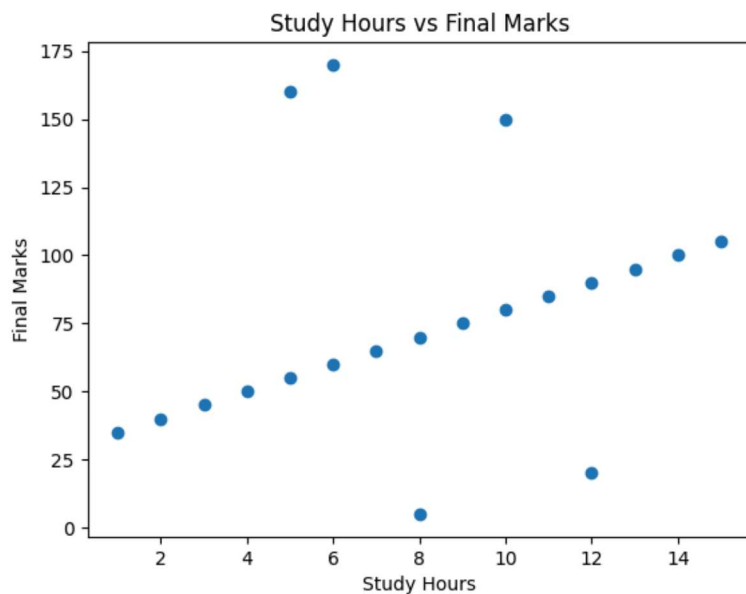
```
Index(['Student_ID', 'Study_Hours', 'Final_Marks'], dtype='object')
Student_ID Study_Hours Final_Marks
0          1           1         35
1          2           2         40
2          3           3         45
3          4           4         50
4          5           5         55
```

Draw Scatter Plot

```
plt.scatter(df["Study_Hours"], df["Final_Marks"])

plt.xlabel("Study Hours")
plt.ylabel("Final Marks")
plt.title("Study Hours vs Final Marks")

plt.show()
```

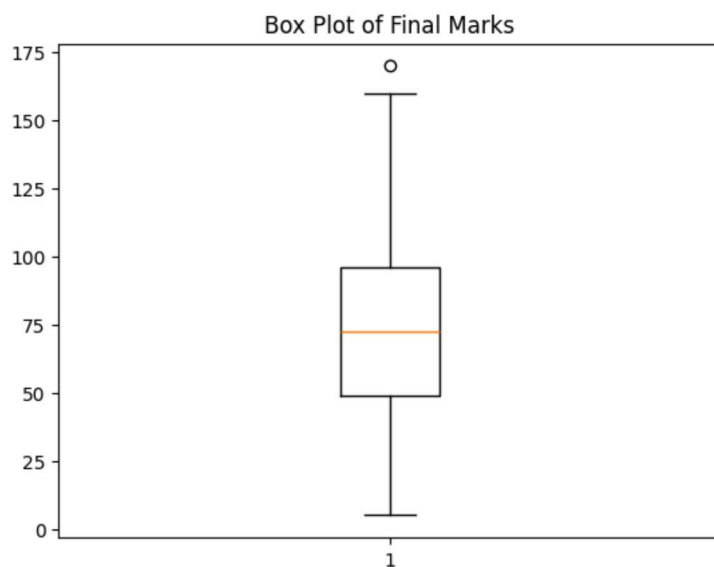


Find Correlation

```
correlation = df["Study_Hours"].corr(df["Final_Marks"])  
  
print("Correlation =", correlation)  
  
Correlation = 0.2591874606161196
```

Detect Outliers (Box Plot)

```
plt.boxplot(df["Final_Marks"])  
  
plt.title("Box Plot of Final Marks")  
  
plt.show()
```



Detect Outliers Using the IQR Method

```
Q1 = df["Final_Marks"].quantile(0.25)  
Q3 = df["Final_Marks"].quantile(0.75)  
  
IQR = Q3 - Q1  
  
lower = Q1 - 1.5 * IQR
```

```
upper = Q3 + 1.5 * IQR

outliers = df[(df["Final_Marks"] < lower) | (df["Final_Marks"] > upper)]

print(outliers)
```

	Student_ID	Study_Hours	Final_Marks
19	20	6	170

```
correlation_matrix = df.corr(numeric_only=True)

print(correlation_matrix)
```

	Student_ID	Study_Hours	Final_Marks
Student_ID	1.000000	0.625832	0.556739
Study_Hours	0.625832	1.000000	0.259187
Final_Marks	0.556739	0.259187	1.000000